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(1) Research Program/Project/Study Title: Banana Lemon and Squash Hair Mask	
Proponent Name:	Nelma P. Delos Santos, MAIED
Academic Rank:	Associate Proffessor III
Co-Authors (with academic rank):	Dr. Ma. Luisa P. Balatayo, Dr. Marybelle F. Amenamen, Prof. Eva B. Cordovero
Campus/Satellite College/Address:	CapSU – Main Campus
Telephone/Cell Phone/Fax Number:	09384992144
E-mail Address:	nelmads1024@gmail.com

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	
Province:	
Region:	

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Agriculture/Fruit and Vegetables	(8) Discipline/Field of Specialization Cosmetics Development
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Banana,lemon and Squash contains many important vitamins and minerals. This fruits and vegetables are main ingredients in making the hair mask. The combination of three would make a product effective and gives more healthy benefits on hair.

The general objective of the study is to develop banana, lemon and squash hair mask, specifically is sought to; 1.determine the qualitative description of banana, lemon and squash hair mask in three treatments in terms of appearance, color, scent, and texture; 2.determine the level of acceptability of banana, lemon and squash hair mask in three (3) treatments in terms of appearance, color, scent and texture; 3.find out the level of effectiveness of banana, lemon and squash hair mask in three (3) treatments in terms of silkiness and smoothing effects when applied to damaged hair; 4.find out if there is a significant difference in the level of acceptability of banana, lemon and squash hair mask in three (3) treatments in terms of appearance, color, scent and texture; 5.find out if there is a significant difference in the level of effectiveness of banana, lemon and squash hair mask in terms of silkiness and smoothing effect when applied to damaged hair; 6.conduct shelf-life testing and chemical analysis of the banana lemon and squash hair mask.

The result of the experiment brought about the following findings:

The qualitative description of banana, lemon and squash hair mask in three treatments in terms of appearance, color, scent, and texture.

Acceptability of banana, lemon and squash hair mask among three treatments in terms of appearance, color, scent and texture. Treatment C described as “VeryAcceptable” Treatment B described as “ Acceptable ” and Treatment A described as “Moderately Acceptable”.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Keywords: Five (5) Point Likert Scale,Qualitative description, and Acceptability

(10) Introduction (This includes the rationale , background of the study, and the objectives which are not divided by subheadings. Incorporate at least (3)reviews)

Hair mask is recorded as hair remedy, to combat baldness with the use of local ingredients that are found surrounding us specifically the fruits and vegetables.

Hair cosmetics is an important tool to help increase adhesion to alopecia and scalp treatments.Hair mask is created to aid the worsen problemas of hair damage caused by chemicals and improper hair practices.

Banana is and edible fruits containing nutritious vitamins and aroma that helps hair tto boost texture, thickness and shine also contains ati microbial properties that heals flaky and dry hair.

Lemon can also be use because of its acid cintent that are useful to reduce dandruff and gives hair shiny appearance.

Squash contains vitamins,minerals and antioxidants beneficial to the skin specially to the sclap when use as a ingredient in making hair mask.

Banana, lemon and squash hair mask is important in preventing damge to the hair by penetrating into the hair and scalp, providing moisture to make hair healthy and decrease hair fall and split ends.

. The general objective of the study is to develop banana, lemon and squash hair mask, specifically is sought to; 1.determine the qualitative description of banana, lemon and squash hair mask in three treatments in terms of appearance, color, scent, and texture; 2.determine the level of acceptability of banana, lemon and squash hair mask in three (3) treatments in terms of appearance, color, scent and texture; 3.find out the level of effectiveness of banana, lemon and squash hair mask in three (3) treatments in terms of silkiness and smoothing effects when applied to damaged hair; 4.find out if there is a significant difference in the level of acceptability of banana, lemon and squash hair mask in three (3) treatments in terms of appearance, color, scent and texture; 5.find out if there is a significant difference in the level of effectiveness of banana, lemon and squash hair mask in terms of silkiness and smoothing effect when applied to damaged hair; 6.conduct shelf-life testing and chemical analysis of the

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banana lemon and squash hair mask.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study used a evaluation sheet as a research instrument. It looked into the four (4) qualitative description such as appearance, color, scent and texture as the variables. In scoring the variables of the study, the Five Point Likert Scale using evaluation sheet was used in evaluating the product.

Statistical Package for Social Sciences (SPSS) was used to generate and processed all the data needed in the study. Mean was used to determine the acceptability of the four qualitative description tested of the products. The ANOVA was used in order to analyze the differences of the products in four qualitative characteristics among three treatments.

Three trials were done to come up with the most acceptable product in terms of qualitative description such as appearance, color, scent, and texture. The finished product was evaluated with the beauty experts for qualitative characteristics and the acceptability was evaluated by the students and consumers.

The treatments used in the product formulation of banana lemon and squash hair mask were the following; Treatment A (80 grams of banana, 80 grams squash, 10ml lemon extract, 10ml mineral oil, 5ml scent and 10ml cethyl alcohol), Treatment B (70 grams of banana, 70 grams squash, 10ml lemon extract, 10ml mineral oil, 5ml scent and 10ml cethyl alcohol, and Treatment C (60 grams of banana, 60 grams squash, 10ml lemon extract, 10ml mineral oil, 5ml scent and 10ml cethyl alcohol).

Product Formulation and Utilization

Procedure in Preparation of Banana

The methods use in the preparation of banana was collected the banana, washed the banana with tap water, peeled the banana using medium size knife, sliced the banana using medium size knife and medium size wooden chopping board, mashed the banana using medium size of bowl and fork, strained the banana using medium size plastic strainer, weighed the banana using medium size weighing scale, packed and labeled.

Procedure in Preparation of Lemon

The methods use in the preparation of lemon was gathered the lemon, washed lemon with tap water, sliced the lemon using medium size knife and medium size wooden chopping board, squeezed the lemon, strained the lemon extract using medium size plastic strainer, measured, packed and labeled.

Procedure in Preparation of Squash

The methods use in the preparation of squash was gathered the squash, washed squash with tap water, sliced the squash using medium size knife and medium size wooden chopping board, peeled squash using medium size peeler, boiled squash in a medium size aluminum casserole, strained the squash using medium size plastic strainer, mashed the squash using fork and plastic mixing bowl, measured, packed and labeled.

Procedure in Preparation of Banana Lemon and Squash Hair Mask

The tools and utensils needed was gathered and prepared. This was treatment A, 10 ml of lemon extract and 80 grams of banana, 80 grams squash, 10 ml scent, 10 ml mineral oil, 5 ml coloring, and 10 grams cetyl alcohol. Treatment B and Treatment C used the same amount of ingredients but it varies only on the amount of banana which uses 70 grams for Treatment B and 60 grams Banana for Treatment C. The three treatments were prepared separately. The mixture was continuously mixed in a mixing bowl until all the ingredients achieved its consistency. The mixtures was packed in plastic container, weighed and labeled based on the treatment. The finished products was divided and one set allowed to stay in a

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room temperature to achieve the desired appearance and its shelf life and the other set was chilled in a refrigerator of the banana lemon and squash hair mask.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Qualitative Characteristics of of banana,lemon and squash hair mask in terms of appearance, scent, and texture

Table 1 shows the qualitative description of Banana, Lemon and Squash Hair Mask in terms of four factors. Treatment C got a highest mean in terms of appearance, got a mean score of 4.60 with a verbal interpretation of “Very Appealing”, color 4.47 with a verbal interpretation of “Very Attractive”, scent 4.47 with a verbal interpretation of “Very Desirable” and texture 4.97 with a verbal interpretation of “Very Fine”. Followed by Treatment B, in terms of appearance a mean score of 4.53, color 4.47, scent 4.13, and texture 4.03. However, Treatment A got the lowest mean score, in terms of appearance a mean score of 4.47, color got a mean score of 4.43, scent got a mean score of 4.13, and texture got a mean score of 3.80. Therefore, this implies that among the three treatments of Banana, Lemon and Squash Hair Mask were found acceptable by the consumers considering the physical properties in terms of appearance, color, scent and texture. However, treatment C was the best in all quality attributes as evaluated by the consumers.

Table 1. Qualitative description of banana, lemon and squash hair mask in terms of appearance, color, scent, and texture.

VARIABLES	Treatment A		Treatment B		Treatment C	
	Mean	VI	Mean	VI	Mean	VI
Appearance	4.47	VA	4.53	VA	4.60	VA
Color	4.43	VA	4.47	VA	4.53	VA
Scent	4.13	D	4.13	D	4.47	VD
Texture	3.80	F	4.03	F	4.97	VF

Legend: VI – Verbal Interpretation

Score	Mean Range	Verbal Interpretation
5	5.00 – 4.20	VA- Very Acceptable
4	4.19 – 3.40	A - Acceptable
3	3.39 – 2.60	MA - Moderately Acceptable
2	2.59 – 1.80	LA - Less Acceptable
1	1.79 – 1.00	NA - Not Acceptable

Determine the acceptability of Banana, Lemon and Squash Hair Mask among three treatments in

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terms of appearance, color, scent and texture

Table 2 shows the qualitative description of Banana, Lemon and Squash Hair Mask in terms of four factors. Treatment C got a highest mean score, in terms of appearance a mean score of 4.60, color got a mean score of 4.47", scent got a mean score of 4.47, and texture got a mean score of 4.97 and an overall mean score of 4.64 with a verbal interpretation of "Very Acceptable". Followed by Treatment B, in terms of appearance a mean score of 4.53, color with a mean score of 4.47, scent got a mean score of 4.13, and texture got a mean score of 4.03 and an overall mean score of 4.29 with a verbal interpretation of "Very Acceptable". However, Treatment A got a lowest mean score, in terms of appearance a mean score of 4.47, color got a mean score of 4.43, scent got a mean score of 4.13, and texture got a mean score of 3.80 with an overall mean score of 4.29 with a verbal interpretation of "Very Acceptable". Therefore, this implies that among the three treatments of Banana, Lemon and Squash Hair Mask were found very acceptable by the consumers considering the physical properties in terms of appearance, color, scent and texture. However, treatment C was the best in all quality attributes as evaluated by the consumers.

Table 2. Acceptability of Banana, Lemon and Squash Hair Mask among three treatments in terms of appearance, color, scent and texture.

VARIABLES	Treatment A		Treatment B		Treatment C	
	Mean	VI	Mean	VI	Mean	VI
Appearance	4.47	VA	4.53	VA	4.60	VA
Color	4.43	VA	4.47	VA	4.53	VA
Scent	4.13	A	4.13	A	4.47	VA
Texture	3.80	A	4.03	A	4.97	VA
General Acceptability	4.21	VA	4.29	VA	4.64	VA

Legend: VI – Verbal Interpretation

Score	Mean Range	Verbal Interpretation
5	5.00 – 4.20	VA- Very Acceptable
4	4.19 – 3.40	A - Acceptable
3	3.39 – 2.60	MA - Moderately Acceptable
2	2.59 – 1.80	LA - Less Acceptable
1	1.79 – 1.00	NA - Not Acceptable

Level of effectiveness of Banana, Lemon and Squash Hair Mask in terms of silkiness and smoothing effects when applied to damaged hair

Table 3 presents the data of the level of effectiveness of Banana, Lemon and Squash Hair Mask in terms of silkiness and smoothing effect when applied by the evaluators. In those factors Treatment A gains highest mean score of 5.0 in silkiness, and 5.0 in moisturizing effect and an overall mean results of 5.0 which interpreted as "Very

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Effective". However, Treatment B was the second highest with the mean score of 4.0 in silkiness, and 4.0 in smoothing effect and an overall mean results of 4.0 which interpreted as "Effective". But, treatment A got a lowest mean score of 3.0 and it was interpreted as "Moderately Effective". This implies that the treatment A of Banana, Lemon and Squash Hair Mask were found as very effective compared to treatment Band C.

Table 3. Level of effectiveness of Banana, Lemon and Squash Hair Mask in terms of silkiness and smoothing effects when applied to damaged hair
Significant difference in the acceptability of Banana,

VARIABLES	Treatment A		Treatment B		Treatment C	
	Mean	VI	Mean	VI	Mean	VI
Silkiness	5.0	VE	4.0	E	3.0	ME
Smoothing Effect	5.0	VE	4.0	E	3.0	ME
Level of Effectiveness	5.0	VE	4.0	E	3.0	ME

Lemon and Squash Hair Mask among three treatments in terms of appearance, color, scent and texture

Table 4 shows the test of difference in the consumers acceptability of the three treatments of Banana, Lemon and Squash Hair Mask among three treatments in terms of appearance, color, scent and texture.

Data revealed that consumers acceptability in terms of appearance (F-value=.412, p-value=.663>.01), color (F-value=.257, p-value=.774>.01) not significantly differ however, in terms of scent (F-value=5.254, p-value=.007<.01), texture (F-value=113.897, p-value=.000<.01) significantly differ. This result implies that consumers have different preferences when it comes to scent and texture produced by the three treatments in favor of treatment C. This result rejects the null hypothesis that consumer's acceptability of scent and texture of Banana, Lemon and Squash Hair Mask tended to be not the same.

Table 4. Difference in the acceptability of Banana, Lemon and Squash Hair Mask among three treatments in terms of appearance, color, scent and texture.

ANOVA

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		Sum of Squ ares	d f	Me an Squ are	F	Si g.	Remarks
APPEAR ANCE	Betw een Grou ps	.267	2	.13 3	.412	.6 6 3	ns
	Withi n Grou ps	28.1 33	8 7	.32 3			
	Total	28.4 00	8 9				
	Betw een Grou ps	.156	2	.07 8	.257	.7 7 4	ns
COLOR	Withi n Grou ps	26.3 00	8 7	.30 2			
	Total	26.4 56	8 9				
	Betw een Grou ps	2.22 2	2	1.1 11	5.25 4	.0 0 7	s
	Withi n Grou ps	18.4 00	8 7	.21 1			
SCENT	Total	20.6 22	8 9				
	Betw een Grou ps	22.8 67	2	11. 433	113. 897	.0 0 0	s
	Withi n Grou ps	8.73 3	8 7	.10 0			
	Total	31.6 00	8 9				

Significant difference in the acceptability of Banana, Lemon and Squash Hair Mask in terms of silkiness and smoothing effects

In this analysis, the table displays the difference in the level of effectiveness of Banana, Lemon and Squash Hair Mask among three treatments in terms of silkiness and smoothing effect when applied. Table 6 shows in terms of all factors used, there were a significant difference among the three treatments because of the p-value of 0 .000 which was lower than the alpha level of significance of 0.01. Therefore, the null hypothesis was rejected that there was a significant difference in the level of effectiveness of Banana, Lemon and Squash Hair Mask among three treatments in terms of silkiness and smoothing effect the results shows that treatment A was very effective compared to treatment B and C when applied by the consumers. This implies that treatment A got the highest result

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compared to Treatment B and C as highly effective as hair mask proportion.

Table 5. Difference in the acceptability of Banana, Lemon and Squash Hair Mask in terms of silkiness and smoothing effect.

		ANOVA					Remarks
		Sum of Squares	df	Mean Square	F	Sig.	
MOISTURIZING	Between Groups	58.022	2	29.011	1305.500	.000	S
	Within Groups	1.933	87	.022			
	Total	59.956	89				
SILKINESS	Between Groups	58.022	2	29.011	1305.500	.000	S
	Within Groups	1.933	87	.022			
	Total	59.956	89				

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Banana, lemon and squash can be used as an added ingredients for added texture in making hair mask.

Fruit and vegetables contains vitamin and minerals that helps the hair to be well nourished that makes the hair smooth and silky.

All treatments of banana, lemon and squash hair mask were all acceptable to all evaluators considering its qualitative descriptions.

Recommendations

Based on the findings and conclusions of the study, the following recommendations were drawn.

Banana Lemon and Squash Hair Mask as an added ingredients in making Hair Mask is recommended.

It is further recommended to the farmers, students, consumers, teachers, entrepreneurs and future researchers to use banana, lemon and squash in making homemade hair mask.

More intensive studies should be conducted along this line, but using different treatments and proportions.

It is recommended that banana lemon and squash hair mask be used instead of just an ordinary hair mask without natural fruits and vegetables like banana lemon and squash.

The unique combination of banana lemon and squash in making hair mask be recommended to the consumers and students because of its natural ingredients.

The conduct of shelf life is done through observation that after the product was made, it was then chilled and it was found out that after five (5) days, the hair mask has molds inside the container .

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

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The output of this study will be the production of a banana lemon and squash hairmask which is more healthy and less expensive compared to the existing commercial hair mask products in the market.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

I would like to thank first the Lord our God who is always the source of my strength and wisdom in the pursuit of this research study. Likewise, I would like to thank my co-researcher Dr. Ma. Luisa P. Balatayo and Dr. Marybelle F. Amenamen and Prof. Eva B. Cordovero, my fellow professors, and evaluators for all the support they have extended throughout the conduct of this study. And also I would like to thank Capiz State University Key Officials, the RDE office, and FFRDC for the funding support extended for the success of this research endeavor. To God be the Glory.

(17) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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